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Applications of quantum computing to optimization

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***Abstract*—The paper presents the basic aspects of the logic behind Quantum Computing and shows its fields of application and the state of the industry, focusing on the special case of the work at Universidad de Montevideo and the startup Quantum-South, born within it: optimization applications for air cargo transport and investment portfolio optimization applications. The paper content has been presented in a plenary session at IEEE CHILECON 2021. It has been intended to motivate professionals and students to work on Quantum Computing in the field of optimization.**

***Keywords*——Quantum computing, Optimization, Cargo.**

I. Introduction

QUANTUM Computing (QC) is a technology for which there is high hope. It is expected to revolutionize many aspects of the way humanity produces and organizes itself.

Experts believe that QC will strongly impact industries as different as utilities management, advertising, pharmaceutical development, healthcare research, supply chain management, autonomous travel data analysis, fraud detection, automotive, and energy [1], [2].

Its formidable computing power will bring new possibilities that transcend conventional computing.

Some problems have different complexity in QC and conventional computing. Some intractable problems become treatable. In fact, there is a type of problem that classical computing cannot solve in a reasonable amount of time, but QC can. This is independent of the usefulness or not of this problem. This has been called Quantum Supremacy and is defined as when a programmable quantum device can solve a problem that no classical computer can solve in any feasible amount of time (irrespective of the usefulness of the problem). Google claimed to reach that goal in 2019, in a paper published in Nature, but IBM expressed skepticism.

On the other hand, Quantum Advantage is a lower and more feasible goal. It refers to the demonstrated and measured success to process a real-world problem faster on a quantum computer than on a classic computer. But quantum computers with enough size and processing capability are to run large real-world problems are still to come. Quantum annealing approach may be an exception which we will address later in this article. Besides, these large quantum computers have special installing conditions. Most of them work close to absolute zero. So most of the machines are installed in manufacturers facilities and accessible through the cloud.

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Quantum Utility is a new concept recently introduced by Mattingley-Scott and Doherty [3] from Quantum Brilliance, a company that pursues the development of room-temperature diamond-based quantum computers. They define quantum utility as a quantum system outperforming classical processors of comparable size, weight, and power in similar environments. Their view is to build smaller quantum computers that complement contemporary high-power computing.

An enormous amount of public funding was announced by many governments around the world to promote the development of quantum technologies and quantum computing [4]. For example

- In the United States, the National Quantum Initiative Act was introduced in 2018, with a five-year budget of over USD1.2 billion with the goal to allow the federal government bodies to accelerate the growth of quantum technologies by collaborating with academic institutions and private industry [5].
- Germany announced in May 2021 that it will spend USD2.4 billion in four years, to support the development of quantum computing and related technologies. Germany's Aerospace Center (DLR) will receive about 890 million US dollars - to team up with industrial companies, medium-sized enterprises, and start-ups [6].

Another important fact around QC is the value it is expected to bring to the market. The Boston Consulting Group estimates that QC could create value for 450 to 850 US billion dollars in the next 15 to 30 years. Part of it – 5 to 10 billion – could be available in the next 3 to 5 years as technology scales up.

It is remarkable the fields in which value will be created:

- Cryptography: $40 to $80 billion
- Optimization: $100 to $220 billion, across Aerospace (flight route optimization), Finances (portfolio optimization and risk management) and Logistics (vehicle routing/network optimization).
- Machine learning: $150 to $220 billion, across Automotive (Automated vehicle), Finance (fraud prevention), and others.
- Simulation: $160 to 330 billion across Aerospace, Automotive, Chemistry, Energy, Finance, battery design, materials design, and Pharma, in a variety of applications.

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II. What is Quantum Computing

Quantum Computing is a new computing paradigm. It uses qubits, instead of bits like "classical computing" does.

In [7] we offered a brief introduction of what QC is and its basis. We will succinctly reproduce it in this section, together with the notion of quantum annealing.

A. The Qubit, Superposition, and Measurement

The qubit is the minimum unit of information that QC uses. The qubit has two base states, which can be noted as two vectors as in (1)

$$\text{State 0: } \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \text{ State 1: } \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{1}$$

Beyond those two base states, the qubit can also be a linear combination (with complex coefficients) of them with norm 1, as in (2). So, in general, a qubit is a vector in $\mathbb{C}^2$.

$$\begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix} = \alpha_0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \alpha_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \tag{2}$$

where α_0 and α_1 are complex numbers and $|\alpha_0|^2 + |\alpha_1|^2 = 1$.

The qubit of equation (2) is in a state of *superposition* of the two base states. That is, it is neither in the state "0" nor state "1", but in a superposition of both states.

On the other hand, when a qubit is measured, the output is one of the two base states. The probability that one or another state is measured, is proportional to the square of the modulus of the coefficient of that state in the linear combination. For instance, when measuring the qubit of equation (2) the possible results are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ with probability $|\alpha_0|^2$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ $|\alpha_0|^2$ with and probability $|\alpha_1|^2$.

Two or more qubits form a system. Those qubits are not independent systems with two independent base states. They are a single system with four base states.

When using n qubits, there are 2^n possible base states. The base states can be linearly combined. The norm of each state describing the system is 1. We can alter or measure one or several qubits of a given system, resulting in a base state or a projection over a sub-space.

B. Gates

One-qubit gates shall transform one qubit from one state to another.

Since one-qubit is a vector in $\mathbb{C}^2$, a gate is a transformation $\mathbb{C}^2 \to \mathbb{C}^2$ (a 2x2 matrix).

The input and output shall have norm 1. So, the transformation shall be unitary. $UU^* = 1$. There are infinite one-qubit gates. We shall remember that in Boolean logic, only four gates can be applied to one bit: identity, not, output 0, and output 1.

Some sample one qubit gates are shown in equations (3) to (5) (Identity, NOT and Hadamard gates).

Identity:

$$\begin{gathered} I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{gathered} \tag{3}$$

Not:

$$\begin{gathered} \text{NOT} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{gathered} \tag{4}$$

Hadamard:

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{5}$$

$$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Hadamard gate is particularly important since, when applied to a state "0", its output is a state of superposition of states "0" and "1".

There are other gates, like SWAP and CNOT, that are applicable to two qubits.

C. Universal Quantum Gate Model

Most of the quantum computers available today are based on the universal quantum gate model. Some of the manufacturers are IBM, IONQ, Atom, Quantum Brilliance, IQM, and Honeywell.

These machines have a certain number of qubits over which many of the above-mentioned gates can be applied in succession. This succession provokes the evolution of the quantum system composed of a set of qubits. This is expected to leave the system at the end in a certain state such that when measured, the result is the solution of the problem.

The main limitation nowadays of the Universal Quantum Gate Model is to maintain qubit stability. This problem grows as the number of qubits increases.

We are in the noisy intermediate-scale quantum (NISQ) era. This implies the use of a low number of qubits, for limited applications, and the use of hybrid algorithms like the

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Variational Quantum Eigensolver (VQE) [8]. VQE allows obtaining the lowest energy level of a system and its eigenstates in an iterative process that implies the use of a quantum computer and a classical optimization algorithm.

As for our experience on Logistics applications, today these kinds of systems have a low capacity and can solve only toy problems.

D. Quantum Annealing

Quantum annealing has been designed to solve classical combinatorial optimization problems.

Optimization problem solving implies the minimization of a cost function.

Many problems of practical importance, however, have cost functions with a large number of local minima, making it extremely difficult for classical algorithms to find the global minimum.

Quantum Annealing implies the physical representation of the combinatorial problem in the quantum system.

Quantum annealing starts from a quantum-mechanical superposition of all possible states (candidate states) with equal weights.

The quantum system evolves to its lowest energy level which is associated with the minimum of the cost function. It takes advantage of quantum tunneling, which facilitates the exploration of the space avoiding stagnating in local minima.

D-Wave, NEC, and Fujitsu are manufacturers that offer the services of a machine with this approach.

These machines can solve QUBO (Quadratic unconstrained binary optimization) problems [9], [10].

The QUBO model is expressed by the optimization problem expressed in (6):

$$y = x^t \mathbf{Q} x \tag{6}$$

where x is a vector of binary decision variables and $\mathbf{Q}$ is a square matrix of constants.

It has applications to many important optimization problems, such as

- Quadratic Assignment Problems
- Capital Budgeting Problems
- Multiple Knapsack Problems
- Task Allocation Problems (distributed computer systems)
- Maximum Diversity Problems
- P-Median Problems
- Asymmetric Assignment Problems
- Symmetric Assignment Problems
- Side Constrained Assignment Problems
- Quadratic Knapsack Problems
- Constraint Satisfaction Problems (CSPs)
- Discrete Tomography Problems
- Set Partitioning Problems
- Set Packing Problems
- Warehouse Location Problems
- Maximum Clique Problems
- Maximum Independent Set Problems
- Maximum Cut Problems
- Graph Coloring Problems
- Number Partitioning Problems
- Linear Ordering Problems
- Clique Partitioning Problems
- SAT problems

These kinds of problems have a large number of applications and have been formulated as a QUBO.

Additionally, quantum annealing machines have been developed faster than universal quantum gate machines, thus allowing working with larger problems.

III. Applications of Quantum Computing to Optimization

Optimization is one of the main uses of QC. Since quantum annealing machines are growing, some real problems can be addressed now, and larger ones will be addressable in the future.

A. Cargo Loading – Universidad de Montevideo and Quantum South

In January 2019, Airbus launched the Airbus Quantum Computing Challenge with five problems. The topics were about all aspects of an aircraft's life: from design to operation, from the quality of movement through the air to the revenue stream of airlines [11].

Our research team worked on Problem number 5: Aircraft Load Optimization [12]. Loading containers to an aircraft has many considerations. For example (i) which containers to load on the aircraft to maximize weight; (ii) there is a maximum weight supported; (iii) a maximum volume has to be respected; (iv) the ordering of the containers affects the center of gravity which subsequently affects the aircraft performance such as fuel burn and speed and ultimately the cost; (v) (iv) the ordering of the containers affect the shear which is limited for the structure of the aircraft. That said, the problem proposed was a simplification of reality.

We were one of the five finalists in the challenge. Our solution from October 2019 was recently presented at [13]. Our approach consisted in translating the problem to a knapsack problem, trying to maximize weight to be transported with two restrictions (volume and weight).

To solve that problem, we used the VQE algorithm in IBM quantum computers available online. We proposed an appropriate Hamiltonian to accomplish the goals.

Due to the success of our proposal, we founded the start-up Quantum South [14] to continue working with these kinds of problems. Universidad de Montevideo and Quantum South are allied to foster the development of quantum computing research, the development of products, and quantum workforce construction.

Since 2019 we continued working on perfecting our algorithms in terms of functionalities and efficiency.

Quantum South works with complex optimization problems

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for cargo in airlines and ships leveraging quantum computing software. The goal is to achieve better results than conventional solutions available today. Our main working line is airline cargo, but we are also working with maritime cargo and financial services.

We already implemented our algorithm in different platforms, both using VQE (in universal quantum gate model machines) or QUBO (in quantum annealing machines).

We worked on the aircraft model A330-200F, which is a very popular cargo aircraft. This model supports different kinds of containers with different shapes and sizes. Besides, different airlines have different configurations in terms of the positions and sizes available for the containers.

Our current solution supports all the above-mentioned restrictions, allowing fast computation of the loading of a real cargo aircraft. It is ready for real scenarios.

We are currently working on other optimization applications detailed below:

- Extension of the container load optimization algorithm to passenger aircraft.
- Extension of the container load optimization algorithm to maritime cargo.
- 3D bin package problem which is a well-known combinatorial problem. Its goal is, given a set of packages, to find the optimal combination and arrangement to maximize the volume occupied.
- Aircraft Climb & Descent Fuel optimization problem, which was the Problem number 1 proposed by Airbus.
- Portfolio optimization in the field of finances.

B. *Some Previous Academic Work*

The field of optimization with QC is having a lot of work.

Vehicle routing is a well-studied field. It is a combinatorial optimization problem, which has NP-hard complexity. This makes promising the approaching with QC.

[15] propose several mathematical formulations, some of them are QUBO based, for vehicle routing with time windows. They make a comparative analysis of their results.

In a similar way, [16] introduces new hybrid algorithms for solving vehicle routing problems. They are also based on QUBO. They tested their algorithms in D-Wave Leap framework. The authors indicate that their experiments had promising results and were able to find solutions similar and better than classical algorithms.

Another field of application is finances. [17] analyzes how QC can be applied to a variety of financial problems such as portfolio optimization, arbitrage opportunities, credit scoring, and others. They consider quantum amplitude estimation, which can result in a quantum speed-up for Monte Carlo sampling.

C. *Manufacturers*

Many manufacturers are highlighting the applications of quantum computing to optimization. D-Wave [18] claims "Unlock the power of practical quantum computing today". The company assures that its customers are building quantum applications for problems as diverse as logistics, portfolio optimization, drug discovery, materials sciences, scheduling, fault detection, traffic congestion, and supply chain management. It also reports some use cases in an automotive assembly line and portfolio optimization.

Fujitsu presents its digital annealer [19] and also claims "Quantum Computing Technology, Available Today". Fujitsu reports real-world use cases in the automotive industry, drug discovery, portfolio optimization, inventory management, and personalized digital marketing.

We shall mention NEC [20] which also is developing machines with the quantum annealing approach. The company expects to launch a quantum annealing machine with all-to-all connected qubits in 2023.

IV. IEEE AND QUANTUM COMPUTING

A final word shall be mentioned about IEEE efforts in this field.

IEEE Quantum [21] is an IEEE Future Directions initiative launched in 2019 that serves as IEEE's leading community for all projects and activities on quantum technologies. Its website holds lots of interesting information resources such as news, podcasts, publications, and conferences.

IEEE Quantum Week [22] is the largest conference on QC. It is held annually, starting in 2020. It offers an enormous program made of tutorials, workshops, technical papers, panels, posters, and keynote speakers. All these events were recorded and are available for any registered attendee. Anyone can register, even now that the conference has been held, to take advantage of these information. It is a good resource for experts and for beginners. Special registration fees are available for certain regions to promote the field.

Many IEEE societies are working in the field of QC. For example, the IEEE Consumer Technology Society has just established a Technical Committee on Quantum in Consumer Technology [23]. The mission and objective of the IEEE CTSoc Quantum in Consumer Technology Technical Committee (QCT TC) are to implement the Field of Interest of the IEEE CTSoc and to assist, support, endorse, corroborate and strengthen IEEE CTSoc conferences, publications, standards, education, future directions, new initiatives, emerging technologies, technical sessions, workshops, and research, and other information exchanges in the area of Quantum computing and related technology solutions. QCT TC is a cluster of engineers, practitioners, technologists, scientists, and researchers from Industry and Academia from around the world.

V. CONCLUSION

This paper has offered a brief introduction to the applications of QC to optimization. We have made an outlook of the companies working on QC and the particular case of Universidad de Montevideo and Quantum South research group. The paper content has been presented as a keynote session at IEEE CHILECON 2021. It has been intended to motivate professionals and students to work on QC.

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